



4. Surviving Without a Hinterland

In 1965, a few months after independence, an economic planner whom the Indian government had seconded to us presented me with a thick volume of his report. I scanned the summary to confirm that his plans were based on a common market with Malaysia. I thanked him, and never read it again. He did not understand that if Malaysia was not willing to have a common market while Singapore was a part of it, it was hardly likely to agree now we were independent. We were stripped of our role as the administrative, commercial and military hub of the British Empire in Southeast Asia. Unless we could find and attach ourselves to a new hinterland, the future was bleak.

A few weeks earlier, I had met Dr Albert Winsemius, our Dutch economic adviser. He painted a grim but not hopeless picture. Because of Confrontation by Indonesia, unemployment had risen. If we continued with no common market with Malaysia and no trade with Indonesia, by the end of 1966 unemployment would exceed 14 per cent. This would mean social unrest. “Singapore is walking on a razor’s edge,” he said. He recommended a common market agreement with Malaysia (which was a non-starter) and a resumption of barter trade with Indonesia. He also advised that we seek more favourable entry for Singapore-manufactured goods into the United States, the United Kingdom, Australia and New Zealand.

Winsemius first came to Singapore in 1960 when he led a United Nations Development Programme (UNDP) mission to advise us on industrialisation. I remember his first report to me in 1961 when he laid two preconditions for Singapore’s success: first, to eliminate the communists who made any economic progress impossible; second, not to remove the statue of Stamford Raffles. To tell me in 1961, when the communist united front was at the height of its power and pulverising the PAP government day after day, that I should eliminate the communists left me speechless as I laughed at the absurdity of his simple solution. To keep Raffles’ statue was easy. My colleagues and I had no desire to rewrite the past and perpetuate ourselves by renaming streets or buildings or putting our faces on postage stamps or currency notes. Winsemius said we would need large-scale technical, managerial, entrepreneurial and marketing knowhow from America and Europe. Investors wanted to see what a new socialist government in Singapore was going to do to the statue of Raffles. Letting it remain would be a symbol of public acceptance of the British heritage and could have a positive effect. I had not looked at it that way, but was quite happy to leave this monument because he was the founder of modern Singapore. If Raffles had not come here in 1819 to establish a trading post, my great-grandfather would not have migrated to Singapore from Dapu county in Guangdong province, southeast China. The British created an emporium that offered him, and many thousands like him, the opportunity to make a better living than in their homeland which was going through turmoil and chaos as the Qing dynasty declined and disintegrated.

Now, in 1965, we too faced a future so bleak that I asked Kim San, then minister of finance, to send a trade delegation from our four chambers of commerce and the manufacturers’ association to Africa on “the off-chance of picking up some business”. The delegation went to East and West African countries, but little trade followed.

After grappling with the problem of unemployment for years since we first took office in 1959, all of us in the cabinet knew that the only way to survive was to industrialise. We had reached the limits of our entrepôt trade. The outlook was a further decline. Confrontation from Indonesia was still on and the Malaysians were determined to bypass Singapore. We cast around for solutions and were willing to try any practical idea that could create jobs and enable us to pay our way. One of our soft drink manufacturers suggested to me that we promote tourism; it was labour-intensive, needing cooks, maids, waiters, laundrymen, dry-cleaners, tour guides, drivers and makers of souvenir handicraft. Best of all, it required little capital. We formed the Singapore Tourist Promotion Board. In 1969, we appointed our film magnate, Runme Shaw of Shaw Brothers, as chairman. He was the right man for the job. He was in the film and entertainment industry and knew all about the packaging and selling of sights and sounds, and how to keep visitors amused while in a strange country. In 1972, I officially lighted up our tourism logo, a “merlion” statue, a lion with a mermaid’s tail, that Shaw built at the mouth of the Singapore River. I did little else for the tourist industry except to speak at the occasional convention of visiting professionals or businessmen. To my relief it did create many jobs and put coins into many empty pockets. It reduced but did not solve the unemployment problem.

For that, we concentrated on getting factories started. Despite our small domestic market of two million, we protected locally assembled cars, refrigerators, air-conditioners, radios, television sets and tape-recorders, in the hope that they would later be partly manufactured locally. We encouraged our own businessmen who set up small factories to manufacture vegetable oils, cosmetics, mosquito coils, hair cream, joss paper and even mothballs! And we were able to attract Hong Kong and Taiwanese investors to build factories for toys, textiles and garments.

It was an unpromising start. The Jurong industrial estate in the west of Singapore was empty in spite of the vast sums we had spent on infrastructure. We had more than our share of failures. Singapore was short of water and too small to tolerate heavy pollution of coastal waters, yet the Economic Development Board (EDB) went into a joint venture to recycle paper products, with a businessman who had no manufacturing experience. It also invested in ceramics when we did not have the technical knowhow. Both ventures failed. We had a joint venture with IHI (Ishikawajima-Harima Heavy Industries) in Jurong Shipyard for ship-building and ship-repairing and started to build 14,000 dwt (dead-weight tons) Freedom-type vessels and later 90,000 dwt tankers. But Singapore produced neither steel plates nor engines and had to import them from Japan. After 16 Freedom-type vessels and three tankers, we stopped building ships, except for small vessels of 10,000 dwt. It was not profitable, unlike ship-repairing where the labour content was high.

In the early years any factory was welcome. For example, when I was in London in January 1968 to discuss the British withdrawal, Marcus Sieff, the chairman of Marks & Spencer's, met me at my London hotel. He had seen me on BBC television. He suggested that as Chinese had nimble fingers, Singapore could go into making fish-hooks and lures for trout fishing. This was high-value work because the feathers had to be skilfully attached to the hooks. There were other such products which did not require much capital equipment but created many jobs. His retail network could help market the goods. I must have looked forlorn on television for him to have taken the time to see me. I thanked him but nothing came of it. Not long after, a Norwegian manufacturer of fish-hooks, Mustad, set up a factory in Singapore, employing several hundred workers to make millions of fish-hooks of all shapes and sizes, though not with feathers for trout fishing.



The loss of British military expenditure between 1968 and 1971 was a blow to our economy. It was some 20 per cent of our GDP, providing over 30,000 jobs in direct employment and another 40,000 in support services. I was determined that our attitude to British aid, indeed any aid, should be the opposite of Malta's. When I visited Malta in 1967 to see how it had sorted out its problems after the run-down of the British forces, I was astounded. The Suez Canal had been closed as a result of the Arab-Israeli Six-Day War three months earlier, in June. Ships were no longer going through the Canal, hence the dockyard in Malta was closed, but dock workers on full pay were playing water-polo in the dry dock which they had filled with water! I was shaken by their aid dependency, banking on continuing charity from the British. The British had given fairly generous redundancy payments, including five weeks' salary for each year of service, and had also covered the cost of three months' retraining in Maltese government institutions. This nurtured a sense of dependency, not a spirit of self-reliance.

Healey in 1967 had promised "significant aid" to offset the loss from the run-down of British forces. I was convinced our people must never have an aid-dependent mentality. If we were to succeed we had to depend on ourselves. Even before talks on British aid commenced, I had said in Parliament, on 9 September 1967, "There was a thriving Singapore before the bases were built and manned. If we set about it intelligently and in good heart, there will be a bigger and economically more self-reliant Singapore after the bases have been run down." My attitude was that we wanted the British to give us the earliest notice of those facilities, like the naval dockyard, they would find redundant and hand them over to civilian management while still in military use. Next, assistance should provide Singapore with jobs through industries and not make us dependent on perpetual injections of aid. I warned our workers, "The world does not owe us a living. We cannot live by the begging bowl."

Hon Sui Sen, our most capable permanent secretary, drew up a list of British assets that could be converted to civilian use. The British outlined their approach to the 15,000 acres of real estate they occupied, 11 per cent of Singapore's total area. Land to be used for economic or defence purposes would be made available without charge. The Singapore government was to help sell the remaining land on the open market. But in January 1968, before the negotiations were concluded, the British announced their total withdrawal by 1971.

On my return to Singapore that January, I said in a broadcast, "If we were a soft society then we would already have perished. A soft people will vote for those who promised a soft way out, when in truth there is none. There is nothing Singapore gets for free, even our water we pay for. ... There will be a throbbing and humming industrial, commercial and communication centre long after the British have gone." I felt strongly that the people's morale and confidence would be decisive in the coming battle for Singapore's survival.

That February, we formed the Bases Economic Conversion Department with Sui Sen in charge. I placed it under my portfolio in the Prime Minister's Office to give Sui Sen more clout when dealing with other ministries. His job was to retrain and redeploy redundant workers, take possession of land and other assets the British were vacating, put them to the best use, and negotiate mitigatory aid.

It was important we did not generate rancour and friction over the handover of properties or the provision of aid. To do so would undermine confidence, and whatever aid we might get could never make up for the loss of confidence if relations with the British soured. Moreover, I still hoped for some residual presence of British, Australian and New

Zealand forces after 1971. I told the newly arrived British high commissioner, Sir Arthur de la Mare, in February 1968 that Singapore would accept whatever his government gave, but would not press them. I asked that his government leave whatever it had no further use for instead of destroying them as was the practice. This would create goodwill and sustain pro-British sentiments in Singapore.

By March 1968 the discussions concluded with a £50 million aid package to be spent on British goods and services. Of this, 25 per cent were grants, 75 per cent loans. We spent half on development projects and half on British defence equipment. The British agreed to hand over the naval dockyard at Sembawang including two valuable floating docks the Royal Navy could easily have towed away, provided the Singapore government appointed Swan & Hunter as managing agents under a five-year contract. I had met Sir John Hunter when I was in London in June 1967, and again in October when I visited his dockyards on Tyneside after the Labour Party conference in Scarborough. The Americans, who were anxious to keep the naval dockyard viable, sent army and navy teams in January and February to inspect the facilities. In April 1968 Sui Sen told me the United States would test-use Sembawang's ship-repair facilities from April to June 1968 and provide S\$4 to 5 million of business. That was most encouraging.

This conversion of the naval dockyard to civilian use was successful. Swan & Hunter prospered both at Keppel, our civilian dockyard, and at Sembawang. When two five-year contracts ended in 1978, one of their top managers, Neville Watson, stayed on with Sembawang Shipyard Ltd, the company we had formed to run the dockyard. Eventually he became its chief executive. The company prospered and grew to become SembCorp Industries, a conglomerate listed on the Stock Exchange of Singapore.

Blakang Mati (behind death), an island off Singapore's harbour housing a British Gurkha battalion, became "Sentosa" (tranquillity), a tourist resort. Dr Winsemius got me to stop it from becoming a military training area, a casino or an oil refinery, proposals put up by various ministries to the Bases Economic Conversion unit under Sui Sen. I did not need persuading to veto these proposals. Fort Canning, with all its tunnels and bunkers, the British Army headquarters before the Japanese capture of Singapore, has also been preserved, the buildings turned into a clubhouse for leisure and recreation. Seletar military airfield was converted to civilian use for small cargo planes and small commercial aircraft. The RAF Changi air base has been expanded by land reclamation and developed into Changi International Airport with two runways. The Pasir Panjang military complex is now the National University of Singapore campus at Kent Ridge with 26,000 students.

In his quiet, methodical way Sui Sen converted the real estate to economic use, and his EDB staff attracted investors from around the world to set up industries on former British army land. It was our good fortune that the real estate handover started in 1968 and was completed by 1971, before the oil crisis in 1973. A buoyant world economy, with world trade expanding at about 8–10 per cent per annum, made it easier to convert them to civilian use.

The withdrawal was carried out with goodwill on both sides. The 30,000 retrenched workers were absorbed by industries we attracted from abroad. When the withdrawal was completed in 1971 our people were quietly confident. There was no unemployment, and no land or building was left idle or derelict. The single British battalion with a squadron of helicopters, together with the Australian and New Zealand battalions that formed the FPDA, contributed to stability and security.



After I had settled policies to counter the loss of British military spending, in the autumn of 1968 I took a short sabbatical at Harvard. I had been in office for nine years and needed to recharge my batteries, get some fresh ideas and reflect on the future. The Kennedy School of Government made me an honorary fellow and arranged breakfasts, lunches, dinners and seminars for me to meet a host of distinguished scholars. During the exchanges, they sparked off many useful and interesting ideas. I learnt much about American society and economy by reading and talking to Harvard Business School professors like Professor Ray Vernon. Vernon gave me a valuable lesson on the ever-changing nature of technology, industry and markets, and how costs, especially wages in labour-intensive industries, determined profits. That was how Hong Kong entrepreneurs were able to build up such a successful textile and garments industry. They had been nimble, changing their product lines, patterns and designs with changing fashions. It was a never-ending competition against other equally agile and lower-cost producers in Taiwan and South Korea. And their sales promoters were frequently flying to consult their buyers in New York and other big cities. He dispelled my previous belief that industries changed gradually and seldom moved from an advanced country to a less developed one. Reliable and cheap air and sea transport made it possible to move industries into new countries, provided their people were disciplined and trained to work the machines, and there was a stable and efficient government to facilitate the process for foreign entrepreneurs.

On my first official visit to America in October 1967, I recounted to 50 businessmen at a luncheon in Chicago how Singapore had grown from a village of 120 fishermen in 1819 to become a metropolis of two million. This was because

its philosophy was to provide goods and services “cheaper and better than anyone else, or perish”. They responded well because I was not putting my hand out for aid, which they had come to expect of leaders from newly independent countries. I noted their favourable reaction to my “no begging bowl” approach.

In November 1968 I went down to New York to address some 800 top decision-makers at the Economic Club of New York. My hard-headed and realistic analysis of Singapore’s problems and the dangers in the region, especially the war in Vietnam, was well received. I took pains to end on a sober but upbeat note, painting a silver lining on sombre clouds. I answered their difficult questions frankly and directly. Several of the executives wrote to congratulate me, and after that night Chan Chin Bock, our EDB chief in New York, found it much easier to get access to top American executives. Thereafter, every time I visited America, he would arrange for me to meet 20 to 50 executives. The usual format was drinks before lunch or dinner, conversation at the main table with the important CEOs, then a 20-minute speech followed by questions and answers. Chin Bock explained that most American CEOs had no time to visit Singapore, but they wanted to see and assess the man in charge before they set up a factory there. My meetings were productive because Winsemius had told me how their minds worked; his son was working for a large American business consultancy firm and knew exactly how Americans weighed business risks. They looked for political, economic and financial stability, and sound labour relations to make sure that there would be no disruption in production that supplied their customers and subsidiaries around the world.

That December, I met another group of American executives in the Far East American Council. Originally only 100 were to attend. After that Economic Club dinner, word got around that I was worth listening to, and the numbers swelled to 200. In a note to the cabinet I grumbled, “Eating and talking through the meal while conserving energy and not letting myself go and drink in case I lose my sharp cutting edge is quite a strain. It is part of the price to promote American investments.”

After several years of disheartening trial and error, we concluded that Singapore’s best hope lay with the American multinational corporations (MNCs). When the Taiwanese and Hong Kong entrepreneurs came in the 1960s, they brought low technology such as textile and toy manufacturing, labour-intensive but not large-scale. American MNCs brought higher technology in large-scale operations, creating many jobs. They had weight and confidence. They believed that their government was going to stay in Southeast Asia and their businesses were safe from confiscation or war loss.

I gradually crystallised my thoughts and settled on a two-pronged strategy to overcome our disadvantages. The first was to leapfrog the region, as the Israelis had done. This idea sprang from a discussion I had with a UNDP expert who visited Singapore in 1962. In 1964, while on a tour of Africa, I met him again in Malawi. He described to me how the Israelis, faced with a more hostile environment than ours, had found a way around their difficulties by leaping over their Arab neighbours who boycotted them, to trade with Europe and America. Since our neighbours were out to reduce their ties with us, we had to link up with the developed world – America, Europe and Japan – and attract their manufacturers to produce in Singapore and export their products to the developed countries.

The accepted wisdom of development economists at the time was that MNCs were exploiters of cheap land, labour and raw materials. This “dependency school” of economists argued that MNCs continued the colonial pattern of exploitation that left the developing countries selling raw materials to and buying consumer goods from the advanced countries. MNCs controlled technology and consumer preferences and formed alliances with their host governments to exploit the people and keep them down. Third World leaders believed this theory of neo-colonialist exploitation, but Keng Swee and I were not impressed. We had a real-life problem to solve and could not afford to be conscribed by any theory or dogma. Anyway, Singapore had no natural resources for MNCs to exploit. All it had were hardworking people, good basic infrastructure and a government that was determined to be honest and competent. Our duty was to create a livelihood for two million Singaporeans. If MNCs could give our workers employment and teach them technical and engineering skills and management knowhow, we should bring in the MNCs.

The second part of my strategy was to create a First World oasis in a Third World region. This was something Israel could not do because it was at war with its neighbours. If Singapore could establish First World standards in public and personal security, health, education, telecommunications, transportation and services, it would become a base camp for entrepreneurs, engineers, managers and other professionals who had business to do in the region. This meant we had to train our people and equip them to provide First World standards of service. I believed this was possible, that we could re-educate and reorientate our people with the help of schools, trade unions, community centres and social organisations. If the communists in China could eradicate all flies and sparrows, surely we could get our people to change their Third World habits.

We had one simple guiding principle for survival, that Singapore had to be more rugged, better organised and more efficient than others in the region. If we were only as good as our neighbours there was no reason for businesses to be based here. We had to make it possible for investors to operate successfully and profitably in Singapore despite our lack of a domestic market and natural resources.



We had established the Economic Development Board by statute in August 1961. Winsemius had recommended a one-stop agency so that an investor need not deal with a large number of departments and ministries. This agency would sort out all an investor's requirements whether relating to land, power, water or environmental and work safety. For the first few months, the EDB had experts from the UNDP and the International Labour Office to help them. The EDB's main efforts were in investment promotion, concentrating on the four industries Winsemius had named in his report – ship-breaking and repair, metal engineering, chemicals, and electrical equipment and appliances.

Hon Sui Sen was picked by Keng Swee as the first chairman of the EDB and given the choice of the brightest and best of our scholars who had returned from universities in Britain, Canada, Australia and New Zealand. These young men were inspired by Sui Sen, a quiet, outstanding administrator with an amazing ability to get the best out of those who worked for him. He shaped the culture of the EDB – the enthusiasm, the unflagging spirit, the ingenious ways they got around obstacles – to promote investments and create jobs. He made the EDB so successful and large that he had to break off different components of the organisation, turning the industrial estates section into the Jurong Town Corporation and the development finance section into the Development Bank of Singapore (DBS). Both became leaders in their own fields. DBS helped finance our entrepreneurs who needed venture capital because our established banks had no experience outside trade financing and were too conservative and reluctant to lend to would-be manufacturers.

It was hard legwork for our young EDB officers to interest foreign investors in the opportunities in Singapore, to persuade them to send missions here to see for themselves. When Chin Bock first began visiting corporate offices, the CEOs did not even know where Singapore was. He had to point it out on their globes, a little dot at the tip of the Malay peninsula in Southeast Asia. EDB officers would sometimes call on 40 to 50 companies before getting one to visit Singapore. They worked with inexhaustible energy because they felt the survival of Singapore depended upon them. Ngiam Tong Dow, a young EDB director, later permanent secretary of the ministry of trade and industry, remembered what Keng Swee told him, that every time he drove by a school and saw hundreds of children streaming out, he felt downhearted, wondering how to find jobs for them when they left school.

The EDB officers had imbibed the values and attitudes of the ministers, a willingness to learn from others and a readiness to accept assistance from any quarter. They were helped by their English-language education. We had inherited the English language from the British and had adopted it as our common working language. From the able team in the EDB, I later found three cabinet ministers, S. Dhanabalan, Lee Yock Suan and Yeo Cheow Tong. Several EDB officers, including Joe Pillay and Ngiam Tong Dow, became outstanding permanent secretaries. In addition Pillay was chairman of Singapore Airlines where his financial and business skills made it the most profitable airline in Asia, while Ngiam became chairman of the Development Bank of Singapore.



Winsemius played a crucial role as economic adviser, serving for 23 years until 1984. He visited Singapore twice a year, each time for about three weeks. We paid for his air tickets and hotel bills in Singapore but for nothing else. To keep him up to date, Ngiam, his EDB liaison officer, sent him regular reports and daily copies of the *Straits Times*. His practice was to spend his first week in Singapore in discussions with our officials, the next with executives of MNCs and some Singapore companies, and also with NTUC (National Trades Union Congress) leaders. He would submit his report and recommendations to the minister of finance and to me. Then I would have a working lunch with him alone.

The top executives of the MNCs soon appreciated the value of his role and spoke freely to him of their problems: over-regulation by the government, the rising value of the Singapore dollar, too much job hopping, too restrictive a policy on employing foreign workers and so on. Winsemius had a pragmatic, hands-on approach, a good head for figures, and a knack for getting to grips with the basic issues, ignoring the mass of details. Most of all, he was wise and canny. I learnt much from him, especially about how European and American CEOs think and operate.

In between his visits to Singapore, he would meet me whenever my work took me to London, Paris, Brussels or Amsterdam. He had to endure one hardship. He was a chain-smoker and I was allergic to smoke, so for him our working meals were a real deprivation. Whenever possible the lunch or dinner would be alfresco, to let him smoke. He spoke English fluently if ungrammatically and with a heavy Dutch accent. He had a deep, guttural voice, a leathery face with deeply lined forehead and cheeks, horn-rimmed glasses and hair combed straight back. He once told me he could not explain why he felt a great affinity with Sui Sen and me and could only conclude that there was a “congruence of the Calvinistic and Confucian philosophy of life”. Whatever the reason, it was Singapore's good fortune that he enjoyed working with us.



The government played a key role in attracting foreign investments; we built the infrastructure and provided well-planned industrial estates, equity participation in industries, fiscal incentives and export promotion. Most important, we

established good labour relations and sound macroeconomic policies, the fundamentals that enable private enterprise to operate successfully. Our largest infrastructure development was the Jurong industrial estate, which eventually covered 9,000 acres, with roads, sewers, drainage, power, gas and water all laid out. It had a slow start. By 1961, we had issued only 12 pioneer certificates. (During 1963–65, our years in Malaysia, none were issued by the central government in Kuala Lumpur.) As finance minister, Keng Swee used to attend the foundation laying ceremony and later the official opening of the factory, to create two occasions for publicity over one factory. He did this even for the smallest factory employing a mere handful of workers, like the one making mothballs. When Jurong lay largely empty, people called it “Goh’s Folly”, as Keng Swee himself recalled after investments had flowed in. He was not that self-deprecatory when Jurong was barren.

By the end of 1970, however, we had issued 390 pioneer certificates giving investors tax-free status for up to five years, extended to 10 years for those issued after 1975. Jurong was humming with activity. Our break came with a visit by Texas Instruments in October 1968. They wanted to set up a plant to assemble semiconductors, at that time a high-technology product, and were able to start production within 50 days of their decision. Close on their heels came National Semiconductor.

Soon after, their competitor, Hewlett-Packard (HP), sent out a scout. Our EDB officer had worked on him, getting him any information he needed immediately, and would not give up until he agreed to visit Singapore to see for himself. He was as impressed as Texas Instruments. An EDB project officer was assigned to look after his delegation and everything was made convenient and swift. While HP negotiated to acquire a site for their own factory, they decided to lease the top two floors of a six-storey building. The elevator to lift the heavy machinery needed a big transformer for electricity, but there was none in place in time for the visit of Mr Hewlett himself. Rather than have him walk up six flights of stairs, the EDB got a gigantic cable extended from a neighbouring building, and on the day of the visit the elevator worked. Hewlett-Packard invested. These stories went through the boardrooms of the American electronics industry, and other American electronics companies soon followed. During this period, China was in the mad throes of Mao’s Cultural Revolution. Most investors thought Taiwan and Hong Kong too close to China and headed for Singapore. We welcomed everyone, but when we found a big investor with potential for growth, we went out of our way to help him get started.

By the 1970s, glowing reports on Singapore had appeared in American magazines, including *US News & World Report*, *Harper’s* and *Time*. General Electric (GE) set up in 1970 six different facilities for electrical and electronic products, circuit breakers and electric motors. By the late 1970s GE was to become the largest single employer of labour in Singapore. American MNCs laid the foundations for Singapore’s large high-tech electronics industry. Although we did not know it then, the electronics industry was to mop up our unemployment and turn Singapore into a major electronics exporter in the 1980s. From Singapore they were later to expand into Malaysia and Thailand.

Visiting CEOs used to call on me before making investment decisions. I thought the best way to convince them was to ensure that the roads from the airport to their hotel and to my office were neat and spruce, lined with shrubs and trees. When they drove into the Istana domain, they would see right in the heart of the city a green oasis, 90 acres of immaculate rolling lawns and woodland, and nestling between them a nine-hole golf course. Without a word being said, they would know that Singaporeans were competent, disciplined and reliable, a people who would learn the skills they required soon enough. American manufacturing investments soon overtook those of the British, Dutch and Japanese.

We had carried the burden of unemployment from the time we first took office in 1959 – so many young people seeking jobs that were not there. But by 1971, when the British forces left, I felt we had turned a corner. The number of unemployed did not rise although the British had discharged their 30,000 workers and left another 40,000 without jobs, people who had served them. The American electronics companies had generated so many jobs that unemployment was no longer an issue. Then suddenly the Arab oil embargo struck, following the Arab-Israeli war of October 1973. The quadrupling of the price of oil set back the world economy. We urged our people to conserve energy and reduce consumption of fuel and electricity. There was belt-tightening but no hardship. Economic growth slowed down significantly from 13 per cent (1972) to 4 per cent (1975) while inflation rose from 2.1 per cent (1972) to 22 per cent (1974). To my relief, we did not suffer a big loss of jobs; our unemployment rate remained around 4.5 per cent.

After recovery in 1975, we could afford to be more selective. When our EDB officer asked how much longer we had to maintain protective tariffs for the car assembly plant owned by a local company, the finance director of Mercedes Benz said brusquely, “Forever”, because our workers were not as efficient as Germans. We did not hesitate to remove the tariffs and allow the plant to close down. Soon afterwards we also phased out protection for the assembly of refrigerators, air-conditioners, television sets, radios and other consumer electrical and electronic products.

By the late 1970s we had left our old problems of unemployment and lack of investments behind us. The new problem was how to improve the quality of the new investments and with it the education and skill levels of our workers. We had found our new hinterland in America, Europe and Japan. Modern communications and transportation made it possible for us to link up with these once faraway countries.

In 1997 we had nearly 200 American manufacturing companies with over S\$19 billion worth of investments at book value. Not only were they the largest of our foreign investors, they constantly upgraded their technology and products. This reduced their unit labour costs, enabling them to pay higher wages without losing competitiveness.

Japanese investments were modest in the 1960s and '70s, well behind those of the British and the Dutch. I tried hard to get the Japanese interested, but they were not moving in strength into Southeast Asia to manufacture for export. In the 1960s and '70s, the Japanese invested overseas merely to sell in domestic markets and did not invest much in Singapore because of our small market. But the success of the American MNCs later encouraged the Japanese to manufacture in Singapore for export to the United States, then Europe, and only much later to Japan. China opened up in the 1980s and Japanese investments started to trickle in. When the Japanese yen appreciated against all other major currencies as a result of the Plaza Accord in 1985, they relocated their middle technology factories to Taiwan, Korea, Hong Kong and Singapore and their lower technology ones to Indonesia, Thailand and Malaysia. When they discovered that their investments in Asia yielded much higher returns than those in America and Europe, East Asia became their major destination. By the mid-1990s they had become the largest investors in manufacturing in East Asia.

Our earliest investors had been the British. After the British forces withdrew from Singapore, many of their companies also left, following their flag. I had tried hard to get them to invest but they were suffering from a withdrawal syndrome, retreating from empire back to the security of their home base, which for them was then not productive because of trade union problems. Only after Singapore showed it could make the grade did the British come back in earnest in the late 1970s, this time not to process or trade in raw materials, but to manufacture high value-added products such as pharmaceuticals. Beecham Pharmaceuticals set up a technologically advanced operation to manufacture semi-synthetic penicillin for the Asian market, especially Japan.

It was the British, Dutch and French who first came and incorporated these countries into the world economies through their empires. These former imperial powers, however, were slow to adjust to the new trade and investment patterns of the post-colonial era, and left the fields they had ploughed to be sown by the Americans and the Japanese.



Several well-established MNCs in Singapore were victims of worldwide restructuring, technological discoveries or market shifts. One example stands out in my memory. After several years the EDB finally convinced Rollei, the German camera manufacturer, to relocate in Singapore. High German wages had made them uncompetitive. I visited Rollei-Werke in Braunschweig in 1970 just before Rollei started to transfer their entire production to Singapore to manufacture cameras, flash guns, projectors, lenses and shutters, and to produce cameras for other famous German camera brands. Together with the EDB, Rollei set up a centre to train workers in precision mechanics, precision optics, tool-making and electro-mechanics. Rollei (Singapore) made excellent cameras but changes in the market and in technology caused poor sales. Their R&D was in Germany, their production base in Singapore. This led to poor planning and coordination. They concentrated their R&D on the slower-moving, professional photographic equipment area, whereas the Japanese moved into ever simpler cameras with viewfinders and other gadgets like auto-focus and talking range finders, all made possible by the computer-chip the Germans were slow to develop. After 11 years, Rollei, both in Germany and Singapore, went into receivership.

Rollei's failure was a great blow for Singapore because European investors interpreted it as a failure in the transfer of technology from Europeans to Singaporeans. The EDB had a difficult time explaining that Rollei's failure was because of changes in technology and markets. One consolation was that the 4,000 workers trained in precision engineering became a valuable base for the disk drive industry that arrived in Singapore in the 1980s.

The EDB has been our primary agency to attract a steady flow of ever higher value-added investments. This has enabled Singapore to remain competitive in spite of rising wage and other costs. Their officers are still some of the brightest of our graduates, mostly from universities in America, Britain and Europe. The EDB's present chairman, Philip Yeo, is well-known to CEOs of MNCs as energetic and dependable, able to deliver whatever the EDB has promised.



Looking back, I cannot claim that our economic development and industrialisation worked as we had planned. The early plans before separation were made on the assumption of a common market with Malaysia. Guinness had already paid a deposit for a site in Jurong for a brewery when Tan Siew Sin, the Malaysian finance minister, told Alan Lennox-Boyd, the Guinness chairman, that he would not allow even one bottle of stout to be imported. So Lennox-Boyd set up his brewery in Kuala Lumpur and offered to allow us to forfeit his deposit. We returned it. Years later we repaid Tan Siew Sin's compliment when we refused to reduce the import duty on stout from Malaysia. Guinness settled on a Singapore brewery to produce it for them under licence.

We left most of the picking of winners to the MNCs that brought them to Singapore. A few such as ship-repairing, oil-refining and petrochemicals, and banking and finance were picked by the EDB or Sui Sen, our minister of finance, or

myself personally. Our ministry of trade and industry believed there could be breakthroughs in biotechnology, computer products, speciality chemicals and telecommunication equipment and services. When we were unsure how new research and development would turn out, we spread our bets.

Our job was to plan the broad economic objectives and the target periods within which to achieve them. We reviewed these plans regularly and adjusted them as new realities changed the outlook. Infrastructure and the training and education of workers to meet the needs of employers had to be planned years in advance. We did not have a group of readymade entrepreneurs such as Hong Kong gained in the Chinese industrialists and bankers who came fleeing from Shanghai, Canton and other cities when the communists took over. Had we waited for our traders to learn to be industrialists we would have starved. It is absurd for critics to suggest in the 1990s that had we grown our own entrepreneurs we would have been less at the mercy of the rootless MNCs. Even with the experienced talent Hong Kong received in Chinese refugees, its manufacturing technology level is not in the same class as that of the MNCs in Singapore.

The government took the lead by starting new industries such as steel mills (National Iron and Steel Mills) and service industries like a shipping line, Neptune Orient Lines (NOL), and an airline, Singapore Airlines (SIA). Two ministers were outstanding in their versatility. Hon Sui Sen seeded the Development Bank of Singapore, the Insurance Corporation of Singapore and the Singapore Petroleum Company. Goh Keng Swee conceived of our shipping line, NOL, and through the Pakistani government, recruited Captain M.J. Sayeed to start it up. With the help of Sir Lawrence Hartnett, an Australian expert in ordnance production, Keng Swee set up Chartered Industries of Singapore (CIS), a mint and a factory for small ammunition, placed together because both required tight security and good production tooling. With a practical and resourceful executive director, Ong Kah Kok, CIS succeeded. Philip Yeo, a young permanent secretary, and later chairman of the EDB, took over CIS from Ong Kah Kok and added new activities that later spun off into Singapore Technologies, a high-tech company that, among other things, set up wafer fabrication plants in joint venture with top MNCs.

We had to put our faith in our young officers who had integrity, intellect, energy, drive and application but no record of business acumen. Our top scholars had been chosen from the best of each year's crop of students and sent to the top universities in Britain, Canada, Australia, New Zealand, Germany, France, Italy and Japan, and later, when we could afford it, America. We made them our entrepreneurs to start up successful companies like NOL and SIA. I was fearful that these enterprises would result in subsidised and loss-making nationalised corporations as had happened in many new countries. Sui Sen, who knew his young officers, assured me that it was possible to succeed, that they could match our competitors in these businesses. And he had given clear instructions that the enterprises had to be profitable or be shut down. Both Keng Swee and Kim San, with whom I discussed these bold plans, thought it worth the risk, given the dearth of entrepreneurs. I relied on the judgement of Sui Sen who had picked the officers for the jobs. The projects succeeded. As a result many new companies sprang up under the auspices of other ministers and their ministries. When these also were successful, we turned state monopolies like the PUB (Public Utilities Board), the PSA (Port of Singapore Authority) and Singapore Telecom into separate entities, free from ministerial control, to be run as companies, efficient, profitable and competitive.

The key to success was the quality of the men in charge. Not all our top administrators possess business acumen, an intangible gift. Several did. National Iron and Steel Mills with Howe Yoon Chong as chairman, Keppel Corporation with Sim Kee Boon, and Singapore Airlines with Joe Pillay became household names, leading stocks on the main board of the Stock Exchange of Singapore. When SIA was privatised, we had difficulty finding top quality executives to replace Joe Pillay, such was the scarcity of entrepreneurial talent.



If I have to choose one word to explain why Singapore succeeded, it is "confidence". This was what made foreign investors site their factories and refineries here. Within days of the oil crisis in October 1973, I decided to give a clear signal to the oil companies that we did not claim any special privilege over the stocks of oil they held in their Singapore refineries. If we blocked export from those stocks, we would have enough oil for our own consumption for two years, but we would have shown ourselves to be completely undependable. I met the CEOs or managing directors of all the oil refineries – Shell, Mobil, Esso, Singapore Petroleum and British Petroleum on 10 November 1973. I assured them publicly that Singapore would share in any cuts they imposed on the rest of their customers, on the principle of equal misery. Their customers were in countries as far apart as Alaska, Australia, Japan and New Zealand, besides those in the region.

This decision increased international confidence in the Singapore government, that it knew its long-term interest depended on being a reliable place for oil and other business. As a result the oil industry confidently expanded into petrochemicals in the late 1970s. By the 1990s, with a total refining capacity of 1.2 million barrels per day, Singapore had

become the world's third largest oil-refining centre after Houston and Rotterdam, the third largest oil trading centre after New York and London, and the largest fuel oil bunker market in volume terms. Singapore is also a major petrochemical producer.

To overcome the natural doubts of investors from advanced countries over the quality of our workers, I had asked the Japanese, Germans, French and Dutch to set up centres in Singapore with their own instructors to train technicians. Some centres were government-financed, others were jointly formed with corporations like Philips, Rollei and Tata. After 4–6 months of training, these workers, who were trained in a factory-like environment, became familiar with the work systems and cultures of the different nations and were desirable employees. These training institutes became useful points of reference for investors from these countries to check how our workers compared with theirs. They validated the standards of Singapore workers.